SHAIDA MUHAMMAD



MACHINE LEARNING ENGINEER | AI & DEEP LEARNING ENGINEER

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LinkedIn: linkedin.com/in/shaidamuhammad | Portfolio: shaidamuhammad.github.io
GitHub: github.com/ShaidaMuhammad | Also publishing as: Shaida Muhammad | Gender: Male

Machine Learning Engineer | NLP, Generative AI, LLMs, Deep Learning and Applied AI

PROFESSIONAL PROFILE

Dedicated and analytical Machine Learning Engineer with hands-on experience developing, evaluating, and deploying NLP and classification systems, large language models (LLMs), transformer-based models, sentiment analysis, text summarization, information extraction, and conversational AI. Experienced with Meta LLaMA-2, Phi-3.5, BERT, RoBERTa, XLNet, Hugging Face, and RASA, supported by a strong foundation in data engineering, automated data pipelines, and cloud deployment across AWS, Microsoft Azure, and Google Cloud Platform (GCP). Research interests and project work span deep learning, generative AI, human activity recognition, wearable/IoT sensing, and applied predictive modeling. Committed to continuous learning, collaborative engineering, and practical AI solutions.

CORE EXPERTISE

Machine Learning & AI: Supervised learning, classification, model evaluation, synthetic data, hierarchical classification, deep learning, neural networks, CNNs, RNNs. **NLP & Generative AI:** LLMs, transformers, BERT, RoBERTa, XLNet, summarization, paraphrasing, grammar correction, sentiment analysis, information extraction, chatbots. **Data & Deployment:** Data processing pipelines, XML/CSV processing, cloud services, CI/CD, MLOps foundations, scalable ML applications.

PROFESSIONAL EXPERIENCE

Senior Machine Learning Engineer | Love For Data
January 2024 – June 2024 • Rawalpindi, Pakistan

- Worked on large language models (LLMs) for text summarization and information extraction, enhancing content comprehension.
- Implemented text-classification models using Hugging Face transformers (BERT and RoBERTa) to analyze textual data, including sentiment-analysis tasks.

Machine Learning Engineer | Datacrypt
October 2022 – December 2023 • Remote, UAE

- Developed intelligent chatbots using the RASA Framework, improving user interaction and feedback collection.
- Designed and maintained data-automation pipelines for efficient data handling and processing.
- Deployed multiple machine learning applications on AWS, Azure, and GCP for scalability and reliability.

Machine Learning Engineer | QLU.ai
April 2021 – May 2022 • Islamabad, Pakistan

- Enhanced the company product with machine-learning components focused on NLP and user engagement.
- Trained and deployed NLP models for text classification and analysis, including sentiment-analysis models.
- Conducted data-engineering tasks on textual data to improve model performance.

EDUCATION

MS in Computer Science (18 years) | National University of Sciences and Technology (NUST)
September 2019 – 2023 • Islamabad, Pakistan
CGPA: 3.90. Core courses: Advanced Algorithms Analysis, Mathematical Methods for Computing, Theory of Computation, Operating Systems. Elective courses: Machine Learning, Deep Learning, Data Mining, Natural Language Processing (NLP).

M.Sc. in Computer Science (16 years) | University of Peshawar
October 2015 – March 2018 • Peshawar, Pakistan
Marks: 80.58%. Courses: Data Structures, Algorithms, Databases, Networking, C++, E-Commerce, Digital Logic Design, Software Engineering, Compiler, Artificial Intelligence, Operating Systems.

Bachelor of Computer Science (14 years) | Bacha Khan University Charsadda
October 2013 – August 2015 • Charsadda, Pakistan
Courses: Computer Science, Mathematics-A, Physics.

TECHNICAL SKILLS

Programming Languages & Frameworks: Python, Java, C++, Flask, Django, FastAPI

Machine Learning & Deep Learning Libraries: PyTorch, Scikit-learn, NLTK, SciPy, NumPy, Pandas, TensorFlow, Keras, Hugging Face

NLP, Transformers & AI: Natural Language Processing (NLP), LLMs, Meta LLaMA-2, Phi-3.5, BERT, RoBERTa, XLNet, FastText, RASA, text summarization, text classification, sentiment analysis, paraphrasing, text generation

Cloud, Data & Developer Tools: AWS, Azure, Google Cloud Platform (GCP), EC2, S3, AWS Lambda, Azure Virtual Machines, Azure Functions, Blob Storage, Git, GitHub, GitLab, SQL, Linux, GitHub Actions / CI/CD, XML, CSV, lxml

AWARDS & DISTINCTIONS

Stoori da Pakhtunkhwa (Stars of the Pukhtunkhwa) Scholarship

BISE Peshawar | January 2014. Secured 15th position among the Top 20 students from Government Colleges in BISE Peshawar.

Top 1% in GAT General Test

National Testing Service | January 2019. Scored 83 out of 100 marks with a 99.62 percentile. GAT is a requirement for applying for an MS degree in Pakistan.

PROFESSIONAL CERTIFICATIONS — COURSERA

- Natural Language Processing Specialization on Coursera [L9XZX72XDF2A]
- Natural Language Processing with Classification and Vector Spaces [HV2BR2GHE7SU]
- Natural Language Processing with Probabilistic Models [9ZKN4QEUBSDC]
- Natural Language Processing with Sequence Models [Z84QTPRB9C6L]
- Natural Language Processing with Attention Models [7SYPZ5D8DPDN]
- Introduction to Large Language Models [LEKT5FXJHJ56]
- Generative AI with Large Language Models [RBXM49D5MZH4]
- Build Basic Generative Adversarial Networks (GANs) [Z7LPD7T6M3A4]
- Introduction to Self-Driving Cars [5AYN5M4FG8LW]
- What is Data Science? [XDT4LQRWLGNB]
- Python Essentials for MLOps [96WYAFV6J2WJ]
- AI For Everyone [46G53XPMEA6S]
- How Google does Machine Learning [G3HM7DJ4XG5Z]
- Artificial Intelligence on Microsoft Azure [2Q55W93P5JPL]
- Introduction to Deep Learning [UF89T3S2QVQG]
- Neural Networks and Deep Learning [LVB52JUQ9HU7]
- Supervised Machine Learning: Regression and Classification [Y7ECGZDYCVWY]
- Mathematics for Machine Learning: Linear Algebra [RYF7JRYNYMJS]

SELECTED MACHINE LEARNING & AI PROJECTS

Project portfolio descriptions and technologies are retained from the supplied CV and website.

Voice-Based AI Assistant for Elderly

TECHNOLOGIES Phi-3.5; open-source text-to-speech (TTS) and speech-to-text (STT) models

Developed a voice-based AI assistant designed to support elderly individuals by answering queries related to health, medicines, and other relevant topics. Used Phi-3.5 and integrated open-source TTS and STT models into a comprehensive pipeline. The assistant functions similarly to Siri, providing voice-activated responses to help users manage daily needs and inquiries.

Document Summarization System

TECHNOLOGIES Django; PyTorch; Meta LLaMA-2 13B; quantized model

Developed a document-summarization system using Meta LLaMA-2 integrated into a Django web application in a secure, resource-constrained environment. Summarized documents of different lengths and formats, from a few sentences to hundreds of pages. Leveraged a quantized Meta LLaMA-2 13B model and collaborated with Django and DevOps teams to support efficient processing and format-specific summaries.

Domain Classification System

TECHNOLOGIES PyTorch; fine-tuned RoBERTa; synthetic data generation; Pandas

Developed a domain-classification system for a complex document dataset spanning nearly 100 domains. Addressed limited and noisy labeled data by generating synthetic data and fine-tuning RoBERTa base models. Designed a hierarchical system with 16 parent models for broad domain classification and child models for subdomain classification, achieving around 80% accuracy.

Sentiment Analysis System

TECHNOLOGIES PyTorch; fine-tuned RoBERTa; Pandas

Developed a sentiment-analysis model to classify document paragraphs into five categories: negative, extreme negative, neutral, positive, and extreme positive. The system provided detailed sentiment insights to improve understanding and interpretation of textual data.

Data Processing Pipeline System

TECHNOLOGIES Python; NumPy; Pandas; lxml; XML; CSV

Designed and implemented a data-processing pipeline for Azadea Group brands, including Azadea, Decathlon, and Adidas. Processed more than 10 GB of XML files daily and generated multiple CSV and XML outputs. Worked within 8 GB RAM constraints on a Windows laptop by processing data sequentially to avoid memory overload.

Message Health Checker

TECHNOLOGIES PyTorch; Transformers; RoBERTa; NumPy; Pandas

Developed "Message Health Checker," a RoBERTa-based classification system for LinkedIn messages. Categorizes messages into five multi-label categories with five ranks each to assess tone and effectiveness. Helps users craft personalized messages, with the original project description reporting up to 10x engagement improvement and conversion-metric tracking.

Message Assistant

TECHNOLOGIES PyTorch; Scikit-learn; Transformers; BERT; RoBERTa; FastText; NumPy

Developed a multi-class, multi-label text-classification model for job-advertisement messages sent by recruiters to LinkedIn candidates. The classification estimates message quality and supports an AI-based text assistant. This project is a component of QLU.ai's primary product.

Job Description Splitting into Sections

TECHNOLOGIES Python; NLTK

Created a hybrid algorithm to split job descriptions into logical sections. The component supports improved precision and recall for other elements of QLU.ai's primary product.

Job Description Sections Classification

TECHNOLOGIES PyTorch; Hugging Face; Transformers; BERT; RoBERTa; XLNet

Trained transformer-based models such as BERT, RoBERTa, and XLNet to classify each section of a job description. This helper component improves the accuracy of other QLU.ai product components and helps enhance customer trust.

Text Summarization, Paraphrasing, and Grammar Correction

TECHNOLOGIES Hugging Face; PyTorch

Researched pre-trained models with strong accuracy for paraphrasing, text summarization, and grammar correction. These models served as helper components for QLU.ai's message-generation feature.

Conversational AI System (Chatbot)

TECHNOLOGIES RASA Framework

Developed a conversational AI system to handle communication between recruiters and candidates on LinkedIn.

Pokémon Image Classification

TECHNOLOGIES PyTorch; NumPy; Feed-Forward Neural Network (FFNN); Convolutional Neural Network (CNN)

Academic project applying FFNN and CNN models to classify Pokémon images into different categories.

Names Generation

TECHNOLOGIES PyTorch; NLTK; Recurrent Neural Network (RNN)

Academic project: developed an RNN-based text-generation model to generate names.

Generating Meaningful Word Embeddings

TECHNOLOGIES Python; NumPy; SciPy; Skip-gram; CBOW; SVD

Academic project: trained Skip-gram, continuous bag-of-words (CBOW), and singular value decomposition (SVD) models on a custom dataset to generate meaningful word embeddings.

Deployments on Cloud

TECHNOLOGIES AWS; Azure; GCP; EC2; S3; AWS Lambda; Azure Virtual Machines; Azure Functions; Blob Storage; GitHub Actions

Deployed services on AWS, GCP, and Azure. Developed scalable data pipelines using AWS and Azure services, including EC2 instances with external FTP, S3 connected to AWS Lambda for data processing, and Azure Virtual Machines with Azure Functions linked to Blob Storage. Integrated GitHub Actions as a CI/CD pipeline to automate deployment and streamline workflows across both cloud environments.

TEACHING EXPERIENCE

Instructor — 4-Week Python Online Course | **Python FREE Online Ramzan Course 2025**

March 3, 2025 – March 28, 2025 • Remote, Worldwide

- Delivered a four-week online Python programming course from beginner to advanced level during Ramzan.
- Covered foundational through advanced Python topics in live sessions with real-time coding, assignments, and quizzes.
- Used Google Classroom, Google Meet, and Google Chat for content delivery and interaction with students.
- Published slides, code examples, assignments, and quizzes as open-source resources on GitHub.
- Course repository: https://github.com/ShaidaMuhammad/Python-Free-Ramzan-Course-2025

Visiting Faculty — Computer Science | **Government Shaheed Rizwan Sareer Higher Secondary School**

September 2018 – March 2019 • Charsadda, Pakistan

- Taught Computer Science to F.Sc students, covering theory and practical programming.
- Conducted coding-practice sessions in the computer lab; assigned and evaluated assignments and quizzes.
- Managed attendance records and maintained classroom discipline and engagement.

RESEARCH PUBLICATIONS

Published research across machine learning, NLP, wearable sensing, human activity recognition, IoT, and sustainable materials. Publication links are clickable; the accepted ICAAD submission is separated from published work.

Conference Papers

Enhanced Human Activity Recognition Using Inertial Sensor Data from Smart Wearables: A Neural Network Approach with Residual Connections **[Published]**

Authors: Shaida Muhammad, Kiran Hamza, Hamza Ali Imran, and Saad Wazir.
2024 International Conference on Engineering & Computing Technologies (ICECT). IEEE, 2024, pp. 1–6.
Link: DOI: 10.1109/ICECT61618.2024.10581358

Mukhtasir-Khail-Net: An Ultra-Efficient Convolutional Neural Network for Sports Activity Recognition with Wearable Inertial Sensors **[Published]**

Authors: Hamza Ali Imran, Shaida Muhammad, Saad Wazir, Ataul Aziz Ikram, and Obaidullah Arshad.
2024 4th International Conference on Digital Futures and Transformative Technologies (ICoDT2). IEEE, 2024.
Link: DOI: 10.1109/ICoDT262145.2024.10740217

Paying Attention to Human Activities: Recognizing Daily Activities with an Attention-Based Neural Network and Inertial Sensors **[Published]**

Authors: Shaida Muhammad, Hamza Ali Imran, Kiran Hamza, Ataul Aziz Ikram, and Saad Wazir.
2024 5th International Conference on Innovative Computing (ICIC). IEEE, 2024.
Link: DOI: 10.1109/ICIC63915.2024.11116432

Spak-Net: A Lightweight Convolutional Neural Network for Activity Recognition with Wearable Inertial Sensors **[Published]**

Authors: Shaida Muhammad, Hamza Ali Imran, Wasim Zaman, and Kiran Hamza.
2025 International Conference on Communication Technologies (ComTech). IEEE, 2025.
Link: DOI: 10.1109/ComTech65062.2025.11034469

Journal Papers

Enhancing Recruiter Outreach: Predicting Job Ad Quality with Advanced Machine Learning and NLP Models **[Published]**

Authors: Shaida Muhammad.
Artificial Intelligence and Applications, 3(4), 443–458, 2025.
Link: DOI: 10.47852/bonviewAIA52024083
Uses a multi-label NLP framework for recruiter-message quality, including call to action, common ground, credibility, incentives, and personalization; the published study reports a customized BERT model accuracy of 95.67%.

From Steps to Sentiments: Cross-Domain Transfer Learning for Activity-Based Emotion Detection in Wearable IoT Systems **[Published]**

Authors: Hamza Ali Imran, Qaiser Riaz, Kiran Hamza, Shaida Muhammad, and Björn Krüger.
IEEE Internet of Things Journal, 13(10), 21498–21509, 2026.
Link: DOI: 10.1109/JIOT.2026.3666469

Book Chapter

Machine Learning for Sustainable Concrete: Predictive Approaches Using Industrial Waste Materials **[Published]**

Authors: Muhammad Shahid Khan, Muhammad Khubaib, and Shaida Muhammad.
In Intelligent Predictive Systems: AI and Machine Learning in Engineering, edited by Masoomeh Mirrashid, Danial Jahed Armaghani, and Aydin Azizi. Emerging Trends in Mechatronics. Springer Nature Singapore, 2026, pp. 1–28.
Link: Chapter DOI: 10.1007/978-981-95-7766-8_1
Book DOI: https://doi.org/10.1007/978-981-95-7766-8. Previously listed on the earlier CV under the working title "Machine Learning Approaches for Forecasting the Strength of Concrete Incorporating Fly Ash and Marble Slurry Waste."

Accepted / Publication Pending

Metaheuristic-Optimized Support Vector Regression as a Real-Time Surrogate for Ship–Bridge Pier Collision Demand **[Accepted; publication pending]**

Authors: Muhammad Shahid Khan, Muhammad Khubaib, Shaida Muhammad, Hammad Ahmad, and Muhammad Shahzad.
International Conference on AI Across Disciplines (ICAAD 2026), Science & Engineering track. Accepted for oral presentation (online); proceedings and DOI details pending.
Link: Official conference programme
Research summary: Benchmarks metaheuristic-optimized support vector regression using Firefly Algorithm, Grey Wolf Optimization, and Particle Swarm Optimization against Decision Tree and Random Forest baselines. The study uses 50,000 collision scenarios across six input variables, evaluates four demand measures with five-fold cross-validation, and reports sub-millisecond surrogate evaluation. Paper ID 164. Earlier work details were supplied by the author; no DOI or published proceedings are claimed.

RESEARCH & PROFESSIONAL FOCUS

Applied artificial intelligence; machine learning engineering; deep learning; natural language processing; generative AI; large language models; transformer architectures; text classification; sentiment analysis; document summarization; information extraction; conversational AI; synthetic data; hierarchical classification; time- and resource-efficient data pipelines; cloud ML deployment; MLOps foundations; wearable-sensor human activity recognition; IoT-based emotion detection; support vector regression; metaheuristic optimization; predictive modeling; sustainable-materials modeling.